

A Spherical N-Series Area Law for Regular Refinement Families

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June 03, 2026

Abstract

This paper presents a spherical extension of the N-Series geometric cancellation framework. The central claim is that regular, scale-consistent spherical area approximations can exhibit an even-power error expansion under refinement. When this structure is present, fixed N-Series scale operators cancel successive error terms and raise the observed convergence order. Five tests are reported. Spherical caps, chordal spherical triangle patches, latitude-longitude spherical grid cells, and fixed irregular spherical polygon domains all exhibit the clean order ladder 2, 4, 6, 8 for the raw, two-scale, three-scale, and four-scale approximations respectively. A perturbation test then identifies the regularity boundary: the raw second-order law remains robust under smooth mesh perturbation, while higher-order cancellation becomes sensitive to the loss of scale regularity. The result is a geometric high-order area-refinement principle for regular spherical refinement families, not a black-box method for arbitrary irregular meshes.

1 Introduction

The N-Series framework begins from a simple observation: many geometric approximations carry structured asymptotic error expansions. When the leading errors occur in even powers of the scale parameter, approximations at several scales can be combined to cancel those leading terms.

In planar polygonal and perimeter problems, this produces the order ladder

$$2, \quad 4, \quad 6, \quad 8, \dots$$

The purpose of this paper is to show that the same structure appears in spherical area approximation.

The tests considered here are deliberately elementary but geometrically distinct:

- a spherical cap approximated by a geodesic boundary polygon;
- a right spherical triangle approximated by a chordal triangulated patch;
- a latitude-longitude spherical cell approximated by a chordal grid;
- fixed irregular spherical polygon domains approximated by regular chordal fan refinements;
- a perturbed latitude-longitude grid used to test robustness.

The first four tests confirm the clean N-Series order ladder for regular refinement families, even when the domain boundary is irregular. The fifth test shows that high-order cancellation has a real regularity boundary when the refinement mesh itself is perturbed. This distinction is important: the method should be stated as a high-order principle for regular, scale-consistent refinement families, not as a universal correction for arbitrary meshes.

The novelty of the present paper is not a new local quadrature formula for spherical area. The contribution is the identification of a scale-structured cancellation law for spherical geometric approximations. When the same spherical area approximation is evaluated on a regular refinement family at the scales $N, 2N, 4N, \dots$, the leading even-power error terms can be cancelled by fixed N-Series weights. This separates the local construction of the approximation from the global cancellation mechanism, and explains why the same order ladder appears across caps, spherical triangle patches, latitude-longitude cells, and irregular spherical polygon domains.

2 Main Contributions

This paper makes four contributions.

First, it formulates a spherical N-Series area law for regular refinement families:

$$A_N(D) = A(D) + \frac{C_2(D)}{N^2} + \frac{C_4(D)}{N^4} + \frac{C_6(D)}{N^6} + \dots$$

The statement is conjectural in general, but is strongly supported by the tests reported below.

Second, it shows that the fixed N-Series operators produce the expected successive order improvements in several spherical settings:

$$O(N^{-2}) \longrightarrow O(N^{-4}) \longrightarrow O(N^{-6}) \longrightarrow O(N^{-8}).$$

Third, it separates boundary irregularity from mesh irregularity. Fixed irregular spherical polygon domains preserve the full 2, 4, 6, 8 ladder when the refinement is regular and scale-consistent.

Fourth, it identifies a limitation: perturbing the refinement mesh preserves the raw second-order law but can degrade higher-order cancellation. Thus regularity is a structural condition, not a cosmetic assumption.

3 N-Series Scale Operators

Let D be a spherical domain on the unit sphere. Let $A(D)$ denote its exact spherical area, and let $A_N(D)$ denote a geometric area approximation at scale N . The spherical N-Series model is that, for suitable regular refinement families,

$$A_N(D) = A(D) + \frac{C_2(D)}{N^2} + \frac{C_4(D)}{N^4} + \frac{C_6(D)}{N^6} + \dots$$

The coefficients depend on the domain and on the approximation family, but the powers are even.

The two-scale N-Series operator is

$$\hat{A}_N^{(2)}(D) = A_{2N}(D) + \frac{A_{2N}(D) - A_N(D)}{3}.$$

If the expansion above holds, then

$$\hat{A}_N^{(2)}(D) = A(D) + O(N^{-4}).$$

The three-scale operator is

$$\hat{A}_N^{(3)}(D) = \frac{1}{45}A_N(D) - \frac{4}{9}A_{2N}(D) + \frac{64}{45}A_{4N}(D),$$

and gives

$$\hat{A}_N^{(3)}(D) = A(D) + O(N^{-6}).$$

The four-scale operator is

$$\hat{A}_N^{(4)}(D) = -\frac{1}{2835}A_N(D) + \frac{4}{135}A_{2N}(D) - \frac{64}{135}A_{4N}(D) + \frac{4096}{2835}A_{8N}(D),$$

and gives

$$\hat{A}_N^{(4)}(D) = A(D) + O(N^{-8}).$$

Thus the method is not a new local formula for spherical area. It is a scale-cancellation operator acting on a family of geometric approximations.

Relation to classical extrapolation. The displayed weights are the Richardson–Romberg extrapolation weights for scale doubling and an even-power error expansion. Their numerical arithmetic is classical: each weight set sums to one and annihilates the relevant powers $N^{-2}, N^{-4}, \dots$. The contribution studied here is geometric rather than algorithmic. It is the evidence that several regular spherical area families possess the required even-power structure, together with the observed loss of a clean high-order ladder when scale coherence is disturbed.

4 Spherical N-Series Area Law

The numerical evidence motivates the following statement.

Conjecture. Let D be a spherical domain and let $A_N(D)$ be a regular, scale-consistent geometric area approximation at scale N . If the refinement family preserves the same geometric structure across the scales $N, 2N, 4N$, and $8N$, then the area error may admit an expansion of the form

$$A_N(D) - A(D) = \frac{C_2(D)}{N^2} + \frac{C_4(D)}{N^4} + \frac{C_6(D)}{N^6} + \dots$$

Under this even-power expansion, the N-Series operators produce the successive order improvements

$$O(N^{-2}) \longrightarrow O(N^{-4}) \longrightarrow O(N^{-6}) \longrightarrow O(N^{-8}).$$

The word regular is essential. The perturbation study below shows that raw second-order convergence can survive mesh perturbation, but higher-order cancellation is sensitive to the loss of scale regularity.

5 Evidence I: Spherical Cap Area

For a spherical cap on the unit sphere with polar angle α , the exact area is

$$A_{\text{cap}} = 2\pi(1 - \cos \alpha).$$

The cap boundary was approximated by a geodesic polygon with N equally spaced boundary vertices. The spherical polygon area was computed from spherical angle excess.

The final observed orders at $N = 256$ are shown in Table 1.

The cap test confirms the N-Series ladder in the simplest spherical setting: a rotationally symmetric domain with an exact closed-form area.

α	Raw order	Two-scale order	Three-scale order	Four-scale order
15°	1.99989583	3.99999065	6.00093925	7.99993479
30°	1.99996740	4.00041698	6.00000053	7.95025527
60°	2.00016296	4.00014936	6.00003853	7.99880645
75°	2.00023455	4.00028432	6.00028420	8.00026989

Table 1: Observed orders for spherical cap area at $N = 256$.

Why even powers arise for the cap. For the cap test, the even-power structure can be justified directly. Let $h = 2\pi/N$, let $c = \cos \alpha$, and let $s = \sin \alpha$. Joining the north pole to two adjacent polygon vertices produces a spherical triangle whose oriented area is

$$E_\alpha(h) = 2 \arctan \left(\frac{s^2 \sin h}{1 + 2c + c^2 + s^2 \cos h} \right)$$

for sufficiently small positive h . The numerator is odd in h , the denominator is even and positive near $h = 0$, and therefore $E_\alpha(h)$ is an odd analytic function:

$$E_\alpha(h) = a_1 h + a_3 h^3 + a_5 h^5 + \dots$$

The geodesic polygon area is $A_N = N E_\alpha(2\pi/N)$. Since $N = 2\pi/h$, it follows that

$$A_N = 2\pi a_1 + \frac{(2\pi)^3 a_3}{N^2} + \frac{(2\pi)^5 a_5}{N^4} + \dots$$

The constant term is the cap area $2\pi(1 - \cos \alpha)$, so the cap family has an even-power error expansion. This argument is specific to the rotationally symmetric cap construction; the broader spherical law remains conjectural.

Precision note. The 30° four-scale order 7.95025527 is slightly below the values near 8 seen in the other cap cases. At this scale the four-scale residual is very small, so a one-step observed-order quotient is sensitive to working precision and near-cancellation. The value is retained as measured rather than adjusted.

6 Evidence II: Chordal Spherical Triangle Area

The second test uses a right spherical triangle on the unit sphere with vertices

$$A = (0, 0, 1), \quad B = (1, 0, 0), \quad C = (\cos \beta, \sin \beta, 0).$$

For this triangle, the exact spherical area is

$$A_{\text{exact}} = \beta,$$

where β is measured in radians.

The approximation uses a normalized barycentric grid. Each small reference triangle is mapped to the sphere and treated as a Euclidean chordal triangle in $\mathbb{R}^3$. The raw area A_N is the sum of the chordal triangle areas.

At $N = 64$, the observed orders are shown in Table 2.

This confirms that the order ladder is not restricted to boundary polygons. It also appears in a triangulated spherical patch.

β	Raw order	Two-scale order	Three-scale order	Four-scale order
15°	1.99942072	3.99915618	5.99886117	7.99716985
30°	1.99938698	3.99908575	5.99880136	7.99761444
60°	1.99923969	3.99881849	5.99858915	7.99835976
90°	1.99885313	3.99827889	5.99785193	7.99808456

Table 2: Observed orders for chordal spherical triangle area at $N = 64$.

7 Evidence III: Latitude-Longitude Cell Area

For a latitude-longitude cell on the unit sphere,

$$\lambda_0 \leq \lambda \leq \lambda_1, \quad \phi_0 \leq \phi \leq \phi_1,$$

the exact area is

$$A_{\text{cell}} = (\lambda_1 - \lambda_0)(\sin \phi_1 - \sin \phi_0).$$

The approximation uses a uniform latitude-longitude grid. Each parameter cell is split into two chordal triangles in $\mathbb{R}^3$, and the raw area is the sum of the Euclidean triangle areas.

At $N = 64$, the observed orders are shown in Table 3.

Cell	Raw order	Two-scale order	Three-scale order	Four-scale order
equatorial $30^\circ \times 30^\circ$	1.99996836	3.99999637	5.99994223	7.99911861
midlatitude $30^\circ \times 30^\circ$	1.99998486	3.99996146	5.99998582	8.00085300
wide equatorial $60^\circ \times 30^\circ$	1.99992674	3.99992210	6.00000864	8.00036842
northern $15^\circ \times 60^\circ$	1.99994200	3.99996772	5.99996443	8.00007636

Table 3: Observed orders for latitude-longitude spherical cell area at $N = 64$.

This test is the most directly applied regular-grid example. It connects the N-Series area law to grid-cell geometry used in spherical mesh and geodesy settings.

8 Evidence IV: Irregular Spherical Polygon Domains

The preceding tests used highly structured spherical domains: caps, right spherical triangles, and latitude-longitude grid cells. A natural question is whether the N-Series ladder depends on the boundary being geometrically simple. To test this, fixed irregular convex spherical polygons were used.

For one six-vertex example, the polygon vertices were

$$(4^\circ, -8^\circ), \quad (18^\circ, 5^\circ), \quad (15^\circ, 28^\circ), \quad (2^\circ, 42^\circ), \quad (-14^\circ, 30^\circ), \quad (-10^\circ, 4^\circ).$$

Its exact spherical area, computed by spherical angle excess, was

$$A_{\text{exact}} = 0.325062777898240815426301649532.$$

The approximation used a regular chordal fan refinement. A root vertex was chosen, the polygon was split into fan triangles, and each fan triangle was approximated by a normalized barycentric chordal patch at scale N .

The observed orders for this fixed polygon are shown in Table 4.

N	Raw order	Two-scale order	Three-scale order	Four-scale order
16	1.99929248	3.99269841	5.99184034	8.01301620
32	1.99982228	3.99817005	5.99798552	8.00310155
64	1.99995552	3.99954224	5.99949790	8.00076594

Table 4: Observed convergence orders for a fixed irregular spherical polygon under regular chordal fan refinement.

The full ladder is preserved. At $N = 64$, the observed orders are

$$1.99995552, \quad 3.99954224, \quad 5.99949790, \quad 8.00076594.$$

The same test was then repeated across deterministic irregular polygon families with 5, 6, and 8 vertices and seeds 1, 2, and 3. The aggregate final-level observed orders are shown in Table 5.

Vertices	Raw order	Two-scale order	Three-scale order	Four-scale order
5	1.99993053	3.99981653	5.99991832	7.99955672
6	1.99993586	3.99979158	5.99991390	8.00029966
8	2.00001032	4.00140962	5.99925770	7.99912295

Table 5: Aggregate final-level observed orders for deterministic irregular spherical polygon families under regular chordal fan refinement.

This confirms that the irregular-boundary result is not a single-polygon accident. Boundary irregularity alone does not destroy the N-Series ladder when the refinement remains regular and scale-consistent.

9 Evidence V: Mesh Perturbation and Regularity Boundary

The final test perturbs the interior nodes of a latitude-longitude cell. The boundary nodes are fixed, while the interior grid points are displaced by a smooth deterministic field. The perturbation amplitude is measured as a fraction of the local grid spacing. Three seeds were tested at each perturbation level.

The aggregate final-level mean orders are shown in Table 6.

Perturbation	Raw order	Two-scale order	Three-scale order	Four-scale order
0.00	1.99996836	3.99999637	5.99994223	7.99911861
0.02	1.99997529	4.02004815	4.53899093	2.61899119
0.05	2.00001207	3.44420303	5.68235637	2.77050496
0.10	2.00014360	3.89815337	8.94059560	3.59334301
0.20	2.00066838	3.90830154	6.15480240	4.36890149

Table 6: Mean final-level observed orders for perturbed latitude-longitude cell area.

The perturbation table must be interpreted as a diagnostic rather than as a list of stable asymptotic orders. Two conclusions are robust in the tested window. First, the raw approximation remains close to second order. Second, the two-scale operator remains broadly effective, although it is less uniform than on the unperturbed grid.

The higher columns do not support a uniform sixth- or eighth-order claim under perturbation. In particular, the three-scale value 8.94059560 exceeds the nominal order of that operator. Such a value can arise from transient coefficient cancellation, a sign change in the residual, proximity to a numerical floor, or loss of a common asymptotic expansion across scales. It is therefore not evidence of genuine order greater than six. More generally, the nonmonotone higher-order values show that the single-step statistic

$$p_N = \frac{\log(e_N/e_{2N})}{\log 2}$$

is not measuring a stable asymptotic regime for those perturbed cases.

The conservative conclusion is that smooth perturbation preserves the raw second-order behavior in these experiments and often leaves the two-scale correction useful, but the data do not establish a clean higher-order ladder for the perturbed meshes. Regularity and scale coherence are therefore part of the hypothesis of the spherical N-Series area law.

10 Boundary Irregularity Versus Mesh Irregularity

The irregular-polygon and perturbation tests separate two different phenomena.

Boundary irregularity means that the spherical domain itself is not a cap, triangle, or latitude-longitude rectangle. The batch irregular-polygon test shows that this does not destroy the N-Series ladder when the interior refinement is still regular.

Mesh irregularity means that the refinement family itself loses coherent scale structure. The perturbation test shows that this can degrade the higher-order operators, especially the four-scale operator.

Thus the decisive condition is not that the boundary must be simple. The decisive condition is that the refinement family must preserve scale regularity.

Boundary stress sweep and failure mode. A small stress sweep was also performed by varying the number of polygon vertices, the angular radius of the domain, and the boundary jitter. For well-posed irregular polygons, the full N-Series ladder persisted. For example, five-vertex cases at radii 15° and 30° , with jitter levels 0.2 and 0.5, retained observed orders close to

$$2, \quad 4, \quad 6, \quad 8.$$

The same was true for the eight-vertex case at jitter 0.2.

However, the eight-vertex cases with jitter 0.5 did not converge to the same area under the chordal fan approximation. The raw error remained nearly constant under refinement, and the observed orders collapsed to approximately zero. This indicates a geometric pathology in the polygon/fan pairing, such as nonconvexity, orientation failure, self-overlap, or convergence to a different spherical region. This is not a failure of N-Series cancellation on a valid refinement family; it is a failure of the approximation family to target the intended domain.

The stress sweep therefore sharpens the regularity condition: boundary irregularity is allowed, but the domain and refinement must remain geometrically well posed.

11 Limitations

The results in this paper are numerical and should be interpreted as evidence for a geometric area law, not as a proof for arbitrary spherical domains.

The following limitations are important:

- all tests use convex or well-behaved spherical domains;
- the exact areas are available from closed-form formulas or spherical angle excess;
- the successful high-order cases use regular, scale-consistent refinement families;
- the perturbation study shows that high-order cancellation may degrade when mesh regularity is lost;
- no claim is made for arbitrary nonconvex, self-intersecting, adaptive, or highly irregular spherical meshes.

These limitations are part of the claim. The method is a high-order geometric cancellation principle for regular refinement families, not a universal mesh repair algorithm.

12 Relation to the General N-Series Framework

The spherical area law extends the N-Series framework from planar perimeter and chord approximation to curved-surface area. The operator mechanism is unchanged: one computes the same geometric quantity at several scales and combines those values with fixed weights so that the leading error modes vanish.

The result is significant because it shows that N-Series cancellation is not limited to approximating lengths or constants such as π . It also applies to area approximation on curved surfaces when the refinement family preserves sufficient structure across scales.

13 Conclusion

The conservative conclusion is that spherical N-Series cancellation provides a geometric high-order area-refinement method for regular spherical refinement families. Boundary irregularity is allowed, provided the refinement remains regular and scale-coherent. Thus the relevant hypothesis is not smoothness of the boundary alone, but coherence of the entire approximation family: the domain, orientation, subdivision rule, and target area must remain fixed across refinement.

The method should not be claimed as a black-box correction for arbitrary irregular meshes. Within its regularity class, however, the evidence strongly supports a spherical extension of the N-Series geometric cancellation framework.

A Reproducibility Notes

The numerical results in this paper were generated by standalone Python scripts. Each script computes exact spherical reference areas, constructs a geometric approximation at several refinement scales, applies the N-Series operators, and reports observed convergence orders.

A.1 Spherical Cap Test

The spherical cap results were generated by

`spherical_area_nseries_cap_test.py`.

For a cap of polar angle α on the unit sphere, the exact area is

$$A_{\text{cap}} = 2\pi(1 - \cos \alpha).$$

A.2 Chordal Spherical Triangle Test

The spherical triangle results were generated by

`spherical_triangle_nseries_area_test.py`.

The target is a right spherical triangle with exact area

$$A_{\text{exact}} = \beta.$$

A.3 Latitude-Longitude Cell Test

The regular latitude-longitude cell results were generated by

`spherical_latlon_cell_nseries_area_test.py`.

For a unit-sphere cell, the exact area is

$$A_{\text{cell}} = (\lambda_1 - \lambda_0)(\sin \phi_1 - \sin \phi_0).$$

A.4 Irregular Spherical Polygon Tests

The irregular spherical polygon results were generated by

`spherical_irregular_polygon_nseries_test.py`

and

`spherical_irregular_polygon_batch_nseries_test.py`.

The boundary stress sweep was generated by

`spherical_irregular_polygon_stress_sweep_nseries_test.py`.

The exact area is computed by spherical angle excess. The approximation uses regular chordal fan refinement. The stress-sweep script varies the number of vertices, angular radius, and boundary jitter in order to distinguish well-posed irregular domains from pathological polygon/fan pairings.

A.5 Perturbed Latitude-Longitude Cell Test

The perturbation robustness results were generated by

`spherical_perturbed_latlon_nseries_test.py`.

This script perturbs the interior grid nodes by a smooth deterministic field while keeping the boundary fixed.

A.6 Observed Orders

The reported convergence orders are computed from successive errors by

$$p_N = \frac{\log(e_N/e_{2N})}{\log 2},$$

where e_N is the absolute error at scale N .

References

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